Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/16/2004
PI Name:	Matin, A.C. Ph.D.		
Project Title:	Molecular basis of the altered stress response in bacteria under SMG.		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	02/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNA04CC51G		
Performance Goal No.:			
Performance Goal Text:			
	Our (and other's) previous results and investigations conducted during the current period of the grant show that simulated microgravity (SMG) resembles the general stress response (GSR) under normal gravity (NG) conditions in that both result in conferring comprehensive cellular resistance (CCR) on bacteria. The bacterial resistance apparatus, however, appears to operate differently under SMG. Unlike GSR-conferred CCR, only a partial correlation exists between Sigma S levels and CCR, which develops without the induction of central stress resistance proteins, such as DnaK. We hypothesize that by affecting the macromolecular folding pattern, SMG alters cellular regulatory mechanisms. We will therefore examine the effect of SMG on the transcriptional, translational, and posttranslational control of proteins centrally involved in bacterial resistance. The major specific aims of the proposed research that we expect to address are: 1. To investigate transcriptional regulatory factors for selected genes under SMG and NG conditions. The		

Task Description:	genes to be investigated are: rpoS, dnaK and pexB (dpp) (general stress response-related); yhiX (gadX), uniquely expressed under SMG conditions; and ugd (induced by SMG). 2. To study the translational control, especially the role of sRNAs and the folding pattern,of the rpoS mRNA (That encodes Sigma S) under the two gravity conditions. 3. To study the half-life of the proteins listed in 1, and the translational efficiency of their mRNAs under SMG and NG conditions. The regulation of these genes at each level of control mentioned above is incompletely understood also under NG; the results will therefore also be relevant to bacterial biology in NG. 4. To study the interaction of DnaK with other cell proteins under the two gravity conditions using two-hybrid and tandem affinity purification (Tap) tagging systems. 5. To generate a knockout mutant in the yhiX gene and examine its SMG CCR phenotype; the protein it encodes is synthesized uniquely under SMG conditions. To understand the basis of increased bacterial resistance and virulence under microgravity conditions of space to be able to devise counter- measures to combat it. For safety of space travel and residence, it is important to better control bacterial virulence especially since microgravity compromises human immune response.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	SMG conditions resemble low-shear environments on Earth, such as the brush border microvilli of the respiratory, gastrointestinal and urogenital tracts. These are common routes of microbial infection, and therefore studies of the kind we have proposed and conducted during the current period will not only indicate means of improving astronaut safety, but also contribute to a better control of infectious disease on this planet. Furthermore, these studies can provide fundamental insights on the role of folding patterns of messenger RNAs and proteins in the regulation of translational efficiency and proteins stability, respectively.
Task Progress:	Capsule version. In this period we have extended our preliminary findings and have obtained definitive data concerning: a. The effect of SMG on the resistance of E. coli not only in exponential phase of growth but also in the stationary phase. b. Quantification of the effect of SMG on Sigma S levels using two different methods in the two growth phases of this bacterium. In addition, we have determined the effect of SMG on Sigma S synthesis in the two growth phases at all levels of control: transcriptional, translational, and posttranslational. The results raise the strong possibility that SMG directly or indirectly affects folding patterns of molecules involved in Sigma S regulation. A direct testing of this possibility is now a major focus of our studies. Brief narrative description of progress. We clearly demonstrate that growth under SMG conditions increases the resistance of Escherichia coli to two stresses, namely hyperosmosis and acid stress. Since these are mechanistically very different stresses and E. coli acquires increased resistance also to ethanol stress when grown under SMG, we conclude that SMG growth confers general resistance on this bacterium. Further, we show that the SMG-caused increase in resistance in stationary phase comes on top of that conferred by this phase of growth itself, making the bacteria virtually completely resistant to these stresses. Since stationary phase is the more common state of bacterial existence in nature, these results highlight the danger posed to astronauts by SMG-conferred increased bacterial resistance and virulence. The generalized increase in resistance conferred by SMG is reminiscent of the general stress response, which has been extensively studied in conventional growth systems on Earth, and it is known that an increase in the concentration of the alternate sigma factor, Sigma S (product of rpoS gene) orchestrates this response under these conditions. We therefore quantified this sigma factor after growth in SMG in the two growth phases, using two different methods involving Western blotting, and an appropriate lacZ fusion strain. Surprisingly, we found that Sigma S levels were decreased by growth in SMG in exponential phase. Thus, cells exhibiting increased general resistance did so without a parallel increase in the concentration of this sigma factor, which is opposite to what has been found in conventional flask studies under normal gravity conditions. Consistent with this finding, we observed that an isogenic mutant strain in the rpoS gene, missing Sigma S, developed increased resistance upon cultivation under SMG just like the wild type. Further surprises lay in store. In the stationary phase, SMG-grown cells possessed 100% higher Sigma S levels compared to the normal gravity (HARV-grown) controls, and the rpoS mutant (missing Sigma S) failed to show SMG-conferred resistance. Thus, while SMG confers resistance to the two stresses in both growth phases, this resistance is independent of Sigma S in exponential phase but dependent on it in stationary phase. Combined with the finding of other's that this increased resistance develops without the induction of genes and proteins known to be involved in conferring resistance and virulence raises the strong possibility that an as yet undiscovered mechanism of comprehensive cellular resistance operates under simulated microgravity conditions. Sigma S regulation in conventional flask cultures has been shown to occur at transcriptional, translational, as well as post-translational levels, depending on the stress and conditions involved. To explore the molecular basis of the altered growth phase- related relationship of Sigma S levels and function, we determined the transcription rate of the rpoS gene, the translational efficiency of the rpoS mRNA, as well as the turnover and synthesis rates of Sigma S, using the methods and equations described in the manuscripts published (in Journal of Bacteriology) in this project period (see bibliography). Two salient findings have emerged from these studies. One, SMG markedly increases the translational efficiency of rpoS mRNA; and two, it greatly destabilizes the Sigma S protein, particularly in one of the growth phases. (In stationary phase, this destabilization is less marked, but is nevertheless, highly reproducible.) Conventional flask studies have revealed that both of these forms of regulation are affected by the secondary structure of macromolecules. According to these studies, rpoS mRNA translational efficiency depends upon its secondary structure especially in its untranslated region. This region contains a sequence complementary to bases present close to the initiation codon and the resulting folding of the mRNA tends to prevent its binding to the ribosomes. Several small RNAs (sRNA) and proteins can alter this structure. Thus, for example, the sRNA, RprA, because of its own complementarity to the interfering rpoS mRNA strand can relax the secondary structure, permitting translation. Similarly, the Sigma S protein is degraded by the ClpXP protease whose activity is greatly affected by the tertiary structure of its target proteins (see manuscript published in J. Bacteriology cited in Bibliography section for appropriate references). We are thus pursuing the hypothesis that SMG influences the folding pattern of macromolecules. This hypothesis is consistent with the finding that space microgravity conditions facilitate protein crystallization. We are testing this hypothesis by determining the roles of various sRNAs and proteins on the translational efficiency of rpoS mRNA; as well as direct determination of the folding

	of this molecule. These studies will be extended also to the folding pattern of Sigma S protein. We will also study the regulation at various control levels of other proteins mentioned in the Description section to determine how general is the effect of SMG is in altering regulation at various control levels, and whether this is connected to alteration in folding patterns.
Bibliography Type:	Description: (Last Updated: 03/27/2020)
Abstracts for Journals and Proceedings	Lynch S, Bradie EL, Matin A. "Molecular mechanisms underlying low-shear simulated microgravity increased stress resistance in planktonic and biofilm Escherichia coli cultures." NASA Cell Science Conference, Galveston, Tx. Conference Proceedings, NASA Cell Science Conference, February 2005. , Feb-2005
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Journal/Magazine covers	Matin A, Lynch SV. "Cover in press for ASM News, Submitted in response to invitation, in review; publication date is approximate." ASM News, American Society for Microbiology, Washington, D.C. (Submitted in response to invitation, in review; publication date is approximate--April 2005) , Apr-2005
Presentation	Matin, A. "SMG drastically alters regulatory pattern" American Society for Microbiology Meeting Symposium on Microgravity, New Orleans, La May-2004
Presentation	S. Lynch "Effect of microgravity on E. coli resistance and its molecular basis" Stanford Retreat, Stanford Center, Fallen leaf Lake Nov-2004
Presentation	S. Lynch "Effect of microgravity on E. coli resistance and its molecular basis" Stanford Bug Club Dec-2005